$$x_j(t) - x_i(t) - \ell_i > 0, \qquad 0 \le v_i(t) \le v_{\max}, \qquad -b_{\max} \le a_i(t) \le a_{\max}$$

where

$$\begin{array}{lcl} x & = & \text{rear-bumper position} \\ \ell_i & = & \text{the follower's length} \\ j,\ i & = & \text{every leader } j \text{ and follower } i \\ v_{\max} & = & \text{maximum speed} \\ a_{\max},\ b_{\max} & = & \text{acceleration and braking limits} \end{array}$$

my submission, Q1.2: Runtime Invariant Monitor; kinematic bounds after Treiber & Kesting (2013)

$$n_0 = 10$$

$$\text{replicate until} \quad \frac{t_{n-1,1-\alpha/2}\, s_n}{\sqrt{n}} \le \delta\, |\bar{X}_n|$$

where

n_0	=	initial number of replications
n	=	current number of replications
$\bar{X}_n,\ s_n$	=	sample mean and standard deviation
$t_{n-1,1-\alpha/2}$	=	Student-t critical value
δ	=	required relative precision

sequential procedure after Law & Kelton; my quarantine rule (Q1.2)

$$\theta^* = \arg\min_{\theta} \sum_{i} w_i \, M_i(\theta)$$

$$\text{subject to} \quad \mathrm{RIM}(\theta) = \mathrm{PASS}$$

where

θ	=	scenario and model parameters
$M_i(\theta)$	=	normalized error per metric: M_1 NRMSE flow, M_2 NRMSE speed, M_3 NRMSE occupancy, . . .
w_i	=	metric weights, pre-set by the researcher
$\mathrm{RIM}(\theta)$	=	Runtime Invariant Monitor verdict: a hard constraint, so invalid regions of the search space are unreachable, not merely penalized

my proposal: objective and stopping criterion (Q1.4)

$$\min_{\theta} \sum_{t} \left\| y_t^{\mathrm{obs}} - y_t^{\mathrm{sim}}(\theta) \right\|^2$$

where

$$\begin{aligned}
y_t^{\mathrm{obs}} &= \text{observed measurement at time } t \\
y_t^{\mathrm{sim}}(\theta) &= \text{simulated measurement under parameters } \theta \\
\theta &= \text{parameters, tuned once the model structure is chosen}
\end{aligned}$$

standard calibration formulation

$$Y = f(v_f,\ \theta_T,\ r,\ u,\ \phi,\ h)$$

where

Y = evacuation outcome (clearance time, exposed vehicles, queue duration, bottleneck location)

v_f = fire spread speed

θ_T = departure-timing control

r = route-choice regime

u = contraflow plan

ϕ = phasing policy

h = household vehicle demand

my submission, Q2.3: compact description of the I-210/SR-134 experiment

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho V)}{\partial x} = \frac{\partial \rho}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

where

ρ = density

V = speed

$Q = \rho V$ = flow

Lighthill & Whitham (1955); Richards (1956)

$$n_i(t+1) = n_i(t) + y_i(t) - y_{i+1}(t)$$

$$y_i(t) = \min\{\, n_{i-1}(t),\ Q_i(t),\ N_i(t) - n_i(t)\,\}$$

where

$n_i(t)$ = number of vehicles in cell i at time t

$y_i(t)$ = inflow into cell i during the interval after t

$Q_i(t)$ = inflow capacity of cell i

$N_i(t)$ = holding capacity of cell i

Daganzo (1994), Cell Transmission Model

$$V(k) = \begin{cases} V_{\text{free}}, & k < k_{\min}, \\ V_{\min} + (V_{\text{free}} - V_{\min}) \left[1 - \left(\dfrac{k - k_{\min}}{k_{\max} - k_{\min}}\right)^a\right]^b, & k_{\min} \le k \le k_{\max}, \\ V_{\min}, & k > k_{\max} \end{cases}$$

where

$V(k)$	=	speed assigned to the entering vehicle
k	=	density on the running part of the link
V_{free}	=	free-flow speed
$V_{\min}$	=	minimum speed
$k_{\min}$, $k_{\max}$	=	density thresholds
a, b	=	model parameters

Mezzo: Burghout et al. (2006)

$$w_{AB} = \frac{q_A - q_B}{k_A - k_B}$$

where

w_{AB} = shockwave speed

$q_A,\ q_B$ = flows upstream and downstream of the node

$k_A,\ k_B$ = densities upstream and downstream of the node

Mezzo: Burghout et al. (2006)

$$v_{\text{safe}}(s, v_l) = -b\Delta t + \sqrt{b^2\Delta t^2 + v_l^2 + 2b(s - s_0)}$$

$$v(t + \Delta t) = \min\{v(t) + a\Delta t,\ v_0,\ v_{\text{safe}}(s, v_l)\}$$

where

s = gap to the leader

s_0 = minimum gap

v_l = the leader's speed

a = maximum acceleration

b = braking rate

v_0 = desired speed

Δt = microscopic time step

Gipps-style safe-speed rule

$$x_{\text{virtual}}(t_{\text{current}}) = x_{\text{end of micro link}} + v_{\text{virtual}}\,(t_{\text{current}} - t_{\text{exit}})$$

where

t_{current} = current simulation time

t_{exit} = time the last vehicle on that lane exited the micro region

v_{virtual} = speed assigned to that vehicle in the mesoscopic model

Burghout (2004); Burghout et al. (2005)

$$V = \begin{cases} V_{\text{front}}, & t_1 < t_h \le t_2, \\ \alpha V_{\text{desired}} + (1-\alpha) V_{\text{front}}, \quad \alpha = \dfrac{t_h - t_2}{t_3 - t_2}, & t_2 < t_h \le t_3, \\ V_{\text{desired}}, & t_h > t_3 \end{cases}$$

where

V	=	assigned initial speed
V_{front}	=	speed of the leading vehicle in the same lane
V_{desired}	=	the entering vehicle's desired speed
t_h	=	time headway to the leading vehicle
t_1, t_2, t_3	=	thresholds calibrated from field data ($t_h \le t_1$: do not load)

Burghout et al. (2006), multi-regime loading

$$v_i(T_k + \Delta t_\mu) = \min\left\{v_i(T_k) + a_i \Delta t_\mu,\ v_{0,i},\ v_{\mathrm{safe},i}(T_k)\right\}$$

where

T_k	=	the k-th microscopic synchronization time
Δt_μ	=	microscopic time-step length
$a_i,\ v_{0,i}$	=	vehicle i's acceleration limit and desired speed

my notation; mechanism: Burghout (2004, 2005, 2006)

$$T_{k+1} = T_k + \Delta t_\mu, \qquad c_k = (m_k,\ T_k)$$

where

T_k = the k-th microscopic synchronization time

Δt_μ = microscopic time-step length

m_k = message sent at that time

c_k = corresponding external event in Mezzo

my compact notation; mechanism: Burghout (2004, 2005, 2006)

$$t(e) \;\le\; H \;=\; \min_a \big(\mathrm{clock}_a + \mathrm{LA}_a\big)$$

where

$t(e)$ = timestamp of event e

H = safe horizon: no earlier event can still arrive

clock_a = current logical time of feeding process a

LA_a = lookahead promised by process a

conservative synchronization: Chandy–Misra–Bryant; Fujimoto (2000)

$$\text{causality gate:}\quad t(e) \le H, \qquad \text{feasibility gate:}\quad t_h > t_1$$

where

$$\begin{aligned}
t(e) &= \text{timestamp of the boundary event entering the micro window}\\
H &= \text{conservative safe horizon of the meso partitions}\\
t_h &= \text{loading headway at the micro entry}\\
t_1 &= \text{minimum loading threshold (multi-regime rule, slide 36)}
\end{aligned}$$

my proposal: the HPDES boundary rule (Q3.3)

$$t_{\text{exit}} \;\geq\; T_k + \frac{d}{v_0}$$

where

t_{exit}	=	earliest time a vehicle can leave the micro window
T_k	=	current synchronization time
d	=	remaining distance to the window exit
v_0	=	desired (maximum) speed inside the window

my proposal, grounded in the Gipps speed bound

Contents

The agent can	The researcher must
generate scenarios	interpret invariant-preserving anomalies
run validator and monitor checks	decide: discovery, artifact, or defect
return structured reason codes	approve scientific claims
suggest next experiments	

my submission, Q1.2: the decision boundary

Behavioral decision	**Pel's model component**	**AutoTrafficSim implementation**	**What remains uncertain**
Whether households evacuate and when they leave	Dynamic travel demand model	Participation rate and response curve are kept as scenario inputs in the **Scenario DSL** and varied across a literature-grounded range.	Warning response, departure delay, and how compressed the departure curve is after warning.
Where households go	Dynamic trip distribution model	The origin–destination matrix is an explicit input in the **Scenario DSL** using survey-grounded ranges where available.	Exact destinations. A location-specific destination-choice model is a later extension.
Which routes vehicles take and whether they switch	Dynamic traffic assignment model	The **Simulation Layer** starts each vehicle on a pre-trip route and allows en-route switching when conditions change. The **Runtime Invariant Monitor** checks lane integrity and event ordering on the resulting trace.	Information access, partial compliance, and the minimum improvement required before switching.
How households respond to route guidance	Dynamic traffic assignment input	The fraction of vehicles receiving or following route guidance is a scenario variable in the **Scenario DSL**. **Metrics Computation** reports clearance time and exposed vehicles for each fraction.	The level of route-guidance compliance during a wildfire evacuation.
How many vehicles each household adds	Dynamic travel demand input	One-vehicle and multi-vehicle cases are encoded in the **Scenario DSL** using Zhao and Wong's survey ranges.	Household vehicle use and towing demand.

Pel et al. (2012); mapping: this work

Factor	**Low or baseline**	**Middle**	**High or severe**
Fire spread	Slow spread	Medium spread	Fast spread
Departure timing	Early departure	Middle departure	Late departure
Route choice	Pre-selected route	Dynamic rerouting	Hybrid route choice with pre-trip plan and en-route switching
Contraflow	No contraflow	Partial contraflow	Full or extended contraflow
Phased evacuation	No phasing	Phased zones	Full-area evacuation
Vehicle demand	One vehicle per household	Mixed vehicle use	High multi-vehicle demand, including towing

my submission, Q2.3: AutoTrafficSim scenario matrix for the I-210/SR-134 evacuation study

	North LA study area	Mobiliti, full LA
Nodes (intersections)	28,422	1,110,335
Links (road segments)	76,307	2,185,984
Network length	11,796 km	
Study area	≈ 960 km^2	
Scale	$\approx 3\%$ of full LA	2.38B events, <30 s, 1024 cores

measured 2026-07-10, OSMnx / OpenStreetMap drive network; Mobiliti: Chan et al.

Committee comments

numbers are slide numbers in this talk